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Alternating-Layer ciGSLTs: Cuts, Decoherence, Reflection, and a Bisimulation Metric — Context Note II

*A companion to the “Interaction, Cost, and Physical Correspondences” note. Where the first note opened the GR and QFT correspondences as curiosities, this one records a single line of thinking that ties them together: that the complex/real weight distinction is a **reflection** in the exact sense of the rho calculus, that this reflection organises the decoherence story, the phase structure, and a recursive (alternating n-level) construction, and that the General Relativity reading becomes a concrete, real-layer optimal-transport programme once surfaces are treated as processes. As before, nothing here is claimed as theorem; the register is “curious correspondence worth investigating,” in the spirit of Curry–Howard and of Girard’s Geometry of Interaction. Citations introduced here are flagged for verification.*

0. The throughline

One organising idea and one recurring obstruction run through everything below.

The idea. The split between *complex-weighted* dynamics (coherent, reversible, interference-bearing) and *real-weighted* dynamics (classical, irreversible, probabilistic) is not a parameter set on a single calculus. It is a **reflection** between two layers, structurally identical to rho’s reflection between processes and names. The complex layer plays the role of the *quote* (the name); the real layer plays the role of the *process*. Decoherence is the dynamics that crosses this boundary, and it is exactly the cost endofunctor.

The obstruction (the same shape, every time). At each stage, the property that makes an object *well-defined* is the property that *kills its interesting part*:

- built-in decoherence makes the LTS bisimulation-sound — and forbids the sharp Wigner’s friend (§4);
- the interaction constructor’s rigidity makes the reversible/irreversible cut immovable — and that immovability is what the sharp paradox needs to violate (§2, §4);
- the discount factor makes the bisimulation metric well-defined — and pushes the geometry toward an ultrametric with degenerate curvature (§7).

The resolution is the same each time: route the irreversibility / contraction onto the **temporal** monoid ($::$, the clock) and leave the **spatial** / coherent monoid (K , nearness) free. This is the single move that recurs.

1. Feynman diagrams and cut rules: two senses of “cut”

The slogan “Feynman diagrams coincide with cut rules” is right, but “cut” is overloaded, and the precise version is stronger than the slogan.

- The **interaction cut** of the cost paper, $K(K_p(I, C_p), K_e(J, C_e)) \rightsquigarrow K'(C'_p, C'_e)$, is a *redex* — one reduction step, one local contraction $\text{compute}(I, J, C_p, C_e)$.
- The **Gentzen cut** is *composition*: joining two proofs along a shared formula.

A Feynman diagram needs both, mapped to different parts of the diagram:

- **Vertices = interaction cuts.** A vertex is where two introductions meet along matched surfaces and their continuations combine.
- **Internal lines / propagators = Gentzen cuts.** A propagator is a residual produced by one vertex that becomes the surface consumed by another — concretely a **shared (located) surface** $S(I, \cdot)$.

So the defensible statement is: *a Feynman diagram is a morphism built by composing interaction cuts, and evaluating the diagram is cut elimination.* This is the GoI / categorical-quantum-mechanics identity; its sharpest published form is the Baez–Stay **Rosetta Stone**, which tabulates Feynman diagrams, proofs, and programs as arrows in monoidal categories with composition = cut. (Note: Mike Stay is a co-author — the frame is already inside the house.)

The payoff: the internal line is the single locus of everything hard. A propagator is a *shared* surface, i.e. the contended located stack under an AC K . Therefore:

- **No internal lines = a tree** = the λ / rigid regime: each residual consumed once (linearity), surface structural and unforgeable, object-capability ideal. Tree-level = tropical/cost reading = $\hbar \rightarrow 0$ saddle point = classical = where bisimulation is sound.
- **Internal lines / loops = the rho / AC regime:** a surface shared, competition admitted, histories recombine.

Hence **interference, entanglement, the qCCS congruence failure, and the ambient-authority leak** are all “**there is an internal line,**” seen four ways. The “local move, global reach” knot of the first note is not a knot but a *location*: it is the propagator. **Loop order = surface-sharing = departure from the ocap ideal.** In linear-logic terms the multiplicative fragment is tree-level (MLL proof nets are essentially acyclic; the correctness criterion forbids vicious cycles), and the **exponentials** ($!$, replication, recursion) are where loops live.

Two further identifications:

- **The signature requirement is a vertex selection rule.** Needing the matching signature resource to fire a cut is charge conservation at the vertex: the signature is a gauge charge, distinct token kinds are distinct flavours, and the resource-sufficiency condition is the Feynman rule that licenses the vertex.
- **Confluence is microcausality.** The diamond commuting independent redexes is “space-like vertices commute”; the cases where reduction order matters are the *contended* (time-like)

surfaces. The amplitude’s invariance under vertex reordering is confluence; the sum-over-diagrams arises from **non-deterministic** resource choices (the deterministic λ -regime has one diagram, no sum, tree-level, classical).

2. Factoring the interaction DAG: reversible vs irreversible loops

The DAG inside interaction is not “loops” undifferentiated. There are **Feynman loops** (virtual, coherent — the integral one performs to evaluate a vertex) and **control-flow loops** (recursion, replication — a server cycling to its ready state). They separate cleanly, and the factorisation is the key refinement:

All reversible/unitary loops fold inside interaction; irreversible loops need not.

Read as physics this is the **GKSL / Stinespring split realised in syntax**: the generator of an open system factors into a reversible part $-i[H, \cdot]$ and an irreversible dissipator $\mathcal{D}$; equivalently, every channel dilates to *unitary on a larger space, then partial trace*. Here the reversible core is the dilated unitary (the annihilation rule’s internal compute supplying the ancilla), and the **cell-pop is the partial trace** — the net interaction firing is where the which-path bit is written to the consumed stack and the off-diagonals die.

Criterion: net phlogiston around the cycle. Zero net consumption (the stack returns to its configuration — borrow-and-return, the Fock ladder up then down) is reversible and absorbable into a vertex; nonzero net consumption is irreversible and stays at the LTS. This is computable from the trace — the same instrument the first note nominates as the decoherence detector. **The factorisation is read off the metering, not laid on top of it.**

This **corrects the tree-vs-DAG framing**. The obstruction to bisimulation is not loops, it is *reversible* loops. Bisimulation handles irreversible cyclic LTSs natively — coinduction is built for exactly that; a recursive process is a cyclic LTS and bisimulation is its home. What bisimulation cannot represent is coherent recombination with cancellation. So decoherence is the passage $\text{DAG} \rightarrow \text{DAG-with-no-reversible-loops}$, **not** $\text{DAG} \rightarrow \text{tree}$; acyclicity (termination) is a *separate axis* the first note conflated with decoherence.

The linear-logic gloss makes “non-reversible” concrete: **weakening is decoherence** (discarding a resource = dropping the phlo record = the trace-out = “forgetting the stack”). **Contraction** (unbounded reuse) is where irreversible loops come from — the bang, replication, recursion. So: irreversible loops = the exponential; reversible loops = the multiplicative-but-cyclic core; the cell-pop where weakening happens is the boundary. This refines §1’s “internal line = where interference lives”: only *reversible* internal lines carry interference. A net-consumed shared surface is a classical race, not an interference channel. **Interference needs both sharing and reversibility — two axes, not one.**

Why the factorisation is “by construction.” The worry — that the reversible/irreversible cut is a gauge choice (the Heisenberg cut is mobile, the Stinespring dilation non-unique) — is the right instinct for a continuum open system and the *wrong* instinct for a syntactic one. There, interaction is a partition drawn across a state; here, interaction is a **constructor**. K is a node in the term; there is no orbit to quotient because the grammar already printed it. More strongly: the interaction-cut presentation has exactly **one cost-bearing rewrite rule** (the cut), and it is

irreversible (pops a cell, stamps which-path, raises entropy); the only other dynamics is $\equiv$, whose loops are *trivial* (return to the same class having computed and recorded nothing). A reversible loop that *carries content* therefore has nowhere at top level — not $\equiv$ (content-free), not a K -firing (net-consuming) — and must live compute-internal. So “**reversible $\subseteq$ compute-internal**” is a **corollary of “one irreversible rule plus one trivial-loop equivalence,”** and the boundary is immovable not because K is conveniently marked but because the rewrite system leaves no other slot for coherence.

Consequence: §3’s obstruction does not get *solved*, it fails to *arise*. The LTS *is* the relation of K -firings; every transition is irreversible and which-path-recording; all coherent recombination is compute-internal, strictly below the transition relation. Bisimulation-soundness carries no proof obligation.

The one residual question relocates to *existence* of the local presentation: which interactions can be put in interaction-cut shape with compute genuinely **local**? A reversible correlation that cannot be localised inside a single K because it spans vertices is a **non-Markovian** effect — the one thing the by-construction factorisation cannot absorb (the presentation fails to exist, not the boundary wobbles). Markovianity is the existence condition.

3. The typing of weights: a trichotomy, not a dichotomy

Complex-weighted ciGSLTs describe interaction dynamics — the quantum analogue of the annihilation rule. **Real-weighted ciGSLTs describe the classical coarse LTS.** Stated this way, the §3 “obstruction” of the first note was a *type error*: demanding branching-time bisimulation of the layer where complex interference lives is asking the wrong equivalence of the wrong layer. Typed correctly, each equivalence applies where it is sound — amplitude / linear-time semantics at the fine (complex) layer (interference is its *job*, not a defect); branching bisimulation at the coarse (real) layer (decohered by construction, hence sound). The loop/tree reading of §1 is then a *weight assignment*: reversible loops inside interaction carry complex weights; the tree-level coarse LTS carries real ones. **Loop content = interaction-internal content = complex content, one cut.**

But mixed $\neq$ real. Collapsing mixed states with classical probabilities loses the object the RG programme needs. A *classical* distribution is a **diagonal** density matrix — $\mathbb{R}^+$ over a fixed pointer basis, genuinely real-weighted, the coarse layer. A general *mixed* state is non-diagonal: it still carries off-diagonal coherences that are not $\mathbb{R}^+$ weights over any basis. The clean picture is a **trichotomy**:

Regime	Weights	Object
pure / unitary / interference	complex	interaction-internal ciGSLT
mixed / CPTP channel / residual coherence	operator	CPM (Selinger doubling) — <i>the missing middle</i>
classical / stochastic / diagonal ρ	real	the coarse LTS

The decoherence flow is $\text{complex} \xrightarrow{\text{double}} \text{CPM} \xrightarrow{\text{diagonalise}} \text{real}$: pure $\rightarrow$ mixed by tracing in the environment, mixed $\rightarrow$ classical by killing off-diagonals. The two named buckets are the endpoints;

the **partially-decohered** regime lives at the **operator/CPM** level. The scale-invariant fixed point of the annihilation- rule RG flow — the thing the energetics note wants to chase — *cannot* be real-weighted (that is the trivial IR endpoint); a nontrivial self-similar regime that neither fully decoheres nor stays fully coherent is by definition partially coherent, i.e. **operator-weighted**. The most interesting object in the programme is the one the dichotomy has no name for. (CPM is in the Abramsky–Coecke lineage already cited; Coecke–Kissinger’s doubling-then-decoherence hierarchy is the diagrammatic form of exactly this three-level structure.)

The dial becomes a functor. The first note’s “semiring is the dial” treated R as a static choice. The passage *between* levels is the content, and it is not a re-setting of the dial but a **functor**: double-then-decohere (CPM doubling followed by the diagonalising decoherence map). This is the quantum analogue of the cost endofunctor.

Open target. Is there a *quantum cost endofunctor* whose action on a complex-weighted ciGSLT is the CPM doubling and whose coarse-graining (the cell-pop / which-path forget) is the diagonalisation — making open-system evolution a composite of two functors in the cost paper’s own idiom, with the RG flow as its iteration?

The bisimulation spectrum falls out. The complex/fine layer is sound only up to trace (linear-time) — interference masks branch differences; the real/coarse layer is sound up to bisimulation (branching-time) — decoherence has exposed them. So van Glabbeek’s two endpoints are the two ends of the decoherence flow: **trace at the coherent UV end, bisimulation at the decohered IR end**. The RG step *coarse-grains the dynamics* (fewer degrees of freedom) while *refining the sound equivalence* (interference no longer masks distinctions); these are statements about different objects (the dynamics and the equivalence on it), so they do not conflict. What the flow tracks in the *middle* of the spectrum is open — and it is the same open question as the operator-weighted middle.

4. MQ-calculus and Wigner’s friend

The MQ-calculus (Stay & Meredith) is a concrete witness for the factorisation. Its process weights are $\mathbb{R}^+$; the complex structure lives entirely in the *state* sort (superposition, scalar, unitaries), and COMM is the map that consumes it:

$$x!(S) \mid \mathbf{for}(m \leftarrow x)\{P\} \rightsquigarrow \sum_r |\langle r|S\rangle|^2 P\{r/m\},$$

the Born projection $\mathbb{C} \rightarrow \mathbb{R}^+$ applied *in the act of communicating*. So MQ is **not** a complex-weighted ciGSLT: it is the $\mathbb{R}^+$ instance of the graded ciGSLT, carrying a $\mathbb{C}$ -linear state datum, with Born-at-COMM as the coupling. This is exactly **option (b)** of the first note’s §5 (“unitary evolution *beside* the interaction relation, decoherence identified with the interaction step”), now concrete, and it answers the note’s own diagnostic: the unitaries appear as a distinct *state-layer* reduction ($\llbracket U; P \rrbracket = \llbracket P \rrbracket(U\psi)$), not as transitions in the relation bisimulation ranges over — so **decoherence is genuinely built in**, the good case. MQ never exhibits $\mathbb{C}$ -weighted *branching*: coherent evolution (closed, no COMM) **xor** COMM (Born, $\mathbb{R}^+$, branching), never both. That is the precise sense in which it *sidesteps* §3 rather than solving it — and it sidesteps it via the factorisation.

Wigner’s friend. MQ makes cut placement *syntactically explicit*, so the **naive** paradox dissolves: MWI is one program (all unitaries inside one output state, no COMM), Copenhagen another (nested

for, COMM at each), “either may be executed.” Good clarification — but the **sharp** paradox (Frauchiger–Renner; the Local Friendliness no-go of Bong–Wiseman et al.; Brukner) turns on precisely the freedom that dissolution uses. Its force is that the modeller may *not* freely fix the program: Wigner treats the friend’s measurement as a reversible unitary on the whole lab while the friend treats the same event as a collapse, both apply to the *same run*, and they disagree on a *jointly checkable* statistic. MQ cannot host this, for an instructive reason: the property that makes its LTS bisimulation-sound — Born at *every* COMM — is what forbids Wigner from measuring the friend’s lab in a coherence-revealing basis. Built-in decoherence and modelling-the-sharp-paradox are incompatible **in one fixed ciGSLT**.

So the calculus-native statement, stronger than “two valid programs”:

The reversible/irreversible boundary is immovable *within* a single ciGSLT (§2); the sharp Wigner’s friend is the assertion that **no single ciGSLT refines two observers’ incompatible cuts over one run**. The friend’s calculus has a COMM where Wigner’s has a reversible unitary; the paradox is the non-existence of a common refinement, and the Local Friendliness inequalities are the experimentally checkable witness that it fails.

To *analyse* rather than dissolve, one needs either (a) a **reversible COMM** (Wigner un-measuring the friend) — which breaks built-in decoherence and pushes straight into the complex-weighted / linear-time regime; or (b) **reify the cut as data** and test when two cuts agree on traces — which is the decoherence-functional diagonality condition (consistent histories) *and* the Wells / Leifer–Milner–Sewell “experiment = context, bisimulation = congruence” program read across two observers. That these coincide is the payoff, and it ties to the qCCS non-congruence residue (Ying). The continuum gap the draft flags is the obstacle: the Wells construction must be made to yield a *congruence over the C-state continuum* before two cuts can be compared rather than chosen between.

5. Field strength makes the two-level structure a phase

If the internal theory depends on a background occupation, $\tilde{G}(\phi)$, then “has two-level structure” becomes “has it at field strength ϕ ,” and the characterisation turns from a *property* into a **phase**. The Higgs remark is the mechanism, not an aside.

- The **order parameter / condensate is the located stack** $S(I, \cdot)$. Its presence is the broken phase.
- The **broken symmetry is the AC permutation symmetry of K** — the bag relabelling that, unbroken, makes every process interchangeably adjacent to every stack. *That symmetry is ambient authority*. Unbroken (symmetric, AC, no condensate): everything near everything, long-range, no capability discipline. Broken (located condensate present): only matching surfaces interact, short-range, capability discipline. **The object-capability discipline is the symmetry-breaking condensate; ambient authority is the restored (high-energy) phase**. “Local move, global reach” is the symmetric phase of a Higgs-like transition.

Direction (corrected to the physical sign): locality of the dilation = a vertex’s environment is nearby = short-range interactions = *massive* mediators = condensate **on**. So the clean two-level structure is the **broken, massive, short-range, IR phase**; it *fails* in the **symmetric, massless, long-range, UV phase**, where mediators are massless, a vertex’s environment is everywhere, and

the dilation de-localises. This matches the RG arrow: UV coherent/non-local, IR decohered/local. “Only working at certain energy ranges” = “valid below the symmetry-breaking scale.”

- **“Different field strengths at different nearnesses” is the Yukawa mass spectrum:** different signatures couple to the *one* common condensate with signature-dependent strength — different effective nearness-cost, i.e. different mass and range. Not different vevs; different couplings to one vev.
- **The massless residue is the photon:** a signature that does *not* couple to the condensate stays massless, long-range, global, capability-free even in the broken phase — the residual ambient-authority channel that survives the capability discipline. If the analogy holds, **such a global capability must exist** (the unbroken $U(1)$).

Rigour line. All of this earns the word *phase* only if the breaking is **spontaneous** — a phlo-potential over occupations with a degenerate minimum (a Mexican hat), so the condensate forms because the symmetric configuration is unstable, not because a knob was turned. A convex potential gives only *explicit*, parametric ϕ -dependence: a real but duller characterisation (indexed by ϕ , no transition, no critical point). The interesting Higgs version — with a genuine phase boundary, the scale-invariant fixed point, the operator-weighted middle — requires the non-convex potential. **Whether the phlo-potential has that shape is the one thing the energetics framework could actually decide**, and the thing to settle before writing any of this down as more than a correspondence.

6. The alternating-layer recursion (the central structure)

Instead of two levels, **alternating n -level ciGSLTs**, with the same enforcement mechanism as rho.

In rho the alternation (process $\rightarrow$ name $\rightarrow$ process $\rightarrow \dots$, “never two quotes in a row”) is forced by typing: the only sort-changing maps are quote $\widehat{(\cdot)} : \text{Proc} \rightarrow \text{Name}$ and its drop; there is no Name $\rightarrow$ Name former, and $\widehat{\widehat{P}}$ is unformable. The same two facts hold for the weight layers:

- **Same-type layers fuse.** Coherent-after-coherent is coherent ($\mathbb{C} \otimes \mathbb{C}$: unitaries compose to a unitary); classical-after-classical is classical ($\mathbb{R}^+ \otimes \mathbb{R}^+ = \mathbb{R}^+$). Two adjacent complex layers *are* one; two adjacent real layers *are* one.
- **A genuine level boundary therefore requires a change of scalar** — the quote or the drop. **“No @@” is the coherent-composition law** ($\mathbb{C}$ -in- $\mathbb{C}$ fuses) in syntactic dress.

Sort assignment: complex = quote/name, real = process. This puts decoherence in the right place. With complex = quoted name, quote/drop is **lossless reification**, not decoherence: $\widehat{\widehat{C}}$ contains all of the complex ciGSLT C , and $\widehat{C}$ dropped is C , a structural law, no accounting. The **lossy** step is the *real-layer COMM firing* — running a quoted complex interior to a definite outcome — which is exactly where the cost endofunctor lives. Metering the cut *is* recording which name was dropped. So the quantum cost endofunctor (decohere $\circ$ CPM, §3) is the *dynamics* of the real layer (the analogue of COMM), cleanly separate from the *structural* reflection (quote/drop), as COMM is separate from $\equiv$. Complementarity is quantitative: the which-path log is the distinguishability, the un-consumed interiors keep the visibility, Englert bounds their sum.

Anomalies = the two ways same-type layers fail to fuse:

- $\mathbb{R}$ -in- $\mathbb{R}$ that will not glue into one classical process = **contextuality** (no global joint distribution over the sub-statistics);
- $\mathbb{C}$ -in- $\mathbb{C}$ that will not compose into one coherent process = **non-Markovian memory** (coherence reaching across a level it should have been sealed inside).

Hence **clean alternation** $\iff$ **no contextuality at the classical levels + Markovianity at the quantum levels** $\iff$ **the dilation is local-and-self-similar at every level** — §2/§5's two-level criterion, now stated for the whole tower. The two-level case is $n = 2$, one quote; the n -level tower is the RG depth (each quote = one rung of van Glabbeek's ladder + one annihilation-rule coarse-graining); and the reflexive fixed point $n \rightarrow \infty$ is the **all-scales self-similar object** — the scale-invariant fixed point of the decoherence flow, existing only in the anomaly-free regime.

Why this must be option (c), not (b). What makes rho higher-order is that a name carries a *process* (reflection), not a bolted-on data sort; the same reflection is what lets the tower recurse. MQ's complex stratum is a Hilbert-space datum — a name with no process inside — so it truncates at $n = 1$ and cannot climb. The alternating construction is the completion: make the interior a genuine complex-weighted ciGSLT, and rho's reflection gives the tower its recursion.

6a. The bootstrap signature

The minimal base move collapses the tower into one mutual recursion:

$$I^{\text{real}} = \text{ciGSLT}(K^{\text{real}}, K'^{\text{real}}, K_p^{\text{real}}, K_e^{\text{real}}, \text{near}^{\text{real}}, I^{\text{complex}}, \text{Tokens}^{\text{real}}),$$

$$I^{\text{complex}} = \text{ciGSLT}(K^{\text{complex}}, K'^{\text{complex}}, K_p^{\text{complex}}, K_e^{\text{complex}}, \text{near}^{\text{complex}}, I^{\text{real}}, \text{Tokens}^{\text{complex}}).$$

Writing $\Phi_w(-)$ for “the w -weighted ciGSLT whose surfaces are drawn from $(-)$,” this is $I^{\text{real}} = \Phi_{\mathbb{R}^+}(I^{\text{complex}})$, $I^{\text{complex}} = \Phi_{\mathbb{C}}(I^{\text{real}})$, each a fixed point of the two-step composite $\Phi_{\mathbb{R}^+} \circ \Phi_{\mathbb{C}}$ (one quote/drop cycle). It is rho's reflexive object $\text{Name} = \widehat{\text{Proc}}$ with the two sorts coloured by two scalars. The **surface slot** is the *only* parameter pointing at the other layer; everything else is intra-layer.

The cross-drawing fixes each compute, and they come out dual.

- compute_R takes complex surfaces, lands in real: $\mathbb{C} \rightarrow \mathbb{R}^+$, a complex thing read into a classical record — **measurement** (a dagger-effect $A \rightarrow I$).
- compute_C takes real surfaces, lands in complex: $\mathbb{R}^+ \rightarrow \mathbb{C}$, classical data conditioning coherent evolution — **controlled preparation** (a dagger-state $I \rightarrow A$).

Measurement and preparation are dagger-adjoint, so $\text{compute}_R \dashv^\dagger \text{compute}_C$ and **the dagger-compact structure the QFT/CQM reading leaned on is generated by the recursion, not assumed** — the construction supplies its own scalars and its own dagger. The pattern (measure, feed the outcome forward to choose the next coherent operation, evolve, measure) is **adaptive measurement-based computation** (Raussendorf–Briegel one-way style); the reflexive limit is **self-referential MBQC** — a name carrying a measurement-based computation that measures names — which is universal.

The quantum subtlety enters near at the real layer. Real-layer $\text{near}^{\text{real}} : I^{\mathbb{C}} \times I^{\mathbb{C}} \rightarrow \text{Surf}$ matches two *complex* surfaces, so it is defined exactly when they are **co-measurable** — when the corresponding observables commute. Complex-layer $\text{near}^{\text{complex}}$ matches two *real* surfaces (classical control) — total/tame. This *inverts* the naive expectation: unitary evolution is deterministic and composes without obstruction; complementarity, Kochen–Specker, and contextuality are statements

about *measurement* statistics, and they land at the only nontrivial **near**. **Contextuality is the failure-pattern of $\text{near}^{\text{real}}$** — the complex surfaces that refuse to co-measure, the commutativity graph with no consistent global section. The partiality of **near** carried since the capability section *is* the carrier of the quantum content. This demotes “no @@” from an enforced predicate to a **typing**: $\mathbb{C}$ -in- $\mathbb{C}$ at a surface position is simply unformable, as $\widehat{\widehat{P}}$ is; the physics it forbade has moved to where it belongs — the partiality of **near** (contextuality) and the locality of compute-memory (Markovianity), both conditions on *operators*, not syntactic side-conditions.

Tokens straddle the cut. Each layer keeps its *own* token monoid (different semirings — the dial as two grading objects). But a located stack lives at a surface, and a surface of one layer is a term of the other, so the stack that funds a crossing is keyed to the *opposite* layer:

$$S^{\mathbb{R}^+}(I^{\mathbb{C}}, s :: S') \quad (\text{real tokens at a complex surface — measurement / which-path ledger}),$$

$$S^{\mathbb{C}}(I^{\mathbb{R}}, \dots) \quad (\text{complex tokens at a classical surface — preparation}).$$

The resource is real (the $\mathbb{R}^+$ which-path ledger) but spent *at* a complex surface (the coherent interior being measured): **phlogiston funds the climb**, and the located condensate **straddles the level boundary by construction** — label in one layer, denomination in the other — which is why it can be the order parameter for a transition *between* levels (§5).

6b. Three coherence conditions (definition vs. theorem)

The two equations are complete as a *signature*; degenerate instances satisfy them (e.g. $\text{near}^{\text{real}}$ total and both computes reversible — the cosmetic-colouring collapse to plain reflexive rho). What makes stage one a theorem is the signature **plus**:

1. $\text{compute}_R \dashv^\dagger \text{compute}_C$ (the dual pair; the structural reflection $\widehat{(\cdot)}/\sim$ is the unit/counit of this adjunction);
2. $\text{dom}(\text{near}^{\text{real}}) = \text{the commutativity relation on } I^{\mathbb{C}}$ (contextuality has a home, and only that home);
3. located stacks cross as $S^{\mathbb{R}^+}(I^{\mathbb{C}}, -) / S^{\mathbb{C}}(I^{\mathbb{R}}, -)$ (resource funds the climb; condensate straddles the cut).

Pin (2) first — it turns the partiality carried since the capability section into the source of the quantum content, and once fixed, the dagger forces (1) and (3). The remaining first-turn question recurs at the tower level: is “anomaly-free” the right reading of “no @@,” or does the reflection collapse some anomalies on its own the way $\equiv$ does in rho (the above-or-below-the-congruence question)?

7. A bisimulation metric and analogs of the Einstein equations

Surfaces are processes, so they carry a behavioural pseudometric for free (Desharnais–Gupta–Jagadeesan–Panangaden: $d(P, Q) \in [0, 1]$, $d = 0$ iff bisimilar, a Kantorovich lift of the successor-metric to a metric on distributions, the fixed point of a contractive functional). Three adjustments turn it into something that can carry a field equation; one reorients the picture.

Metrise the match defect, not similarity. Bisimilar processes are behaviourally identical and need never interact; a and $\bar{a}$ interact *because* they differ. Nearness for interaction is **duality**, not sameness. Using the dagger of §6, the right quantity is $d(I^\dagger, J)$ — zero on perfect matches, growing

as the match degrades — defined inside a **cost-bounded neighbourhood** of perfect match. The neighbourhood radius is an interaction range, funded by the located stack (the tropical reading): **range and resource are the same dial**.

From pseudometric to curvature. A field equation needs an infinitesimal line element, i.e. a Hessian of d^2 near the diagonal. This needs (i) a *length* metric — d as an infimum of path-costs, the phlo/tropical reading, so finite nearness is geodesic; (ii) the Kantorovich lift with **quadratic** cost W_2 (not W_1): W_1 gives a metric but no Riemannian shadow, W_2 gives **Otto's** formal Riemannian structure on distributions, which makes $d(x, x + dx)^2 \approx g_{\mu\nu} dx^\mu dx^\nu$ meaningful — *rebuild the bisimulation metric with the 2-Kantorovich lift*; (iii) a reference measure — the **phlo distribution** $\phi : \text{Surf} \times \text{Sig} \rightarrow R$, i.e. the stress-energy of the first note. **Geometry from bisimulation- W_2 ; matter from phlo.**

The signature is Lorentzian, and it is the two monoids. Bisimulation distance is symmetric/positive (Riemannian-flavoured); GR is Lorentzian. The cone comes from the *other* monoid. Spatial metric (symmetric, W_2 -bisimulation) = the K monoid; causal order = reduction $\leadsto$, directed and irreversible = the $::$ monoid (the arrow of computation). A **Lorentzian length space** (Kunzinger–Sämann) is a set with a causal relation plus a time-separation τ ; here causal future = reduction-reachable and $\tau(x, y)$ = the *maximal* phlo along causal paths. Note the extremal flip: spatial **near minimises** cost (tropical min), proper time **maximises** (free fall is longest-aging, maximal phlo). **Min over space, max over time** — the two monoids, with the Lorentzian sign already in which extremum each takes. (Confluence = microcausality, §1, is the same statement: space-like configurations commute, time-like ones do not.)

The field equation is a convexity, not a PDE. The space of processes-mod- bisimulation is infinite-dimensional and non-smooth — exactly the synthetic case. Lott–Villani and Sturm define Ricci lower bounds on a metric-measure space by **displacement convexity of entropy along W_2 -geodesics** (curvature from a metric and a measure, no charts). The Lorentzian, matter-coupled version is in hand: Mondino–Suhr give an optimal-transport formulation of the Einstein equations, and McCann's theorem says displacement convexity of Boltzmann entropy *characterises* the strong energy condition of GR (matter curves spacetime). Instantiated:

Einstein's equation = the statement that the phlo measure controls the convexity of phlo-entropy along timelike W_2 -geodesics of the bisimulation metric — the timelike direction being the reduction order, the entropy the Boltzmann entropy of the token distribution. “Stress-energy $\leftrightarrow$ metric” becomes a provable-or-refutable inequality.

The **local** counterpart, for a PDE-shaped object: **Bakry–Émery** — the Γ_2 / Bochner identity of the **generator of cost-metered reduction** (the infinitesimal cost endofunctor as d'Alembertian) gives a local Ricci object, and $\Gamma_2 \geq K \Gamma$ with K set by phlo is the local field equation. This is the GoI / operator-algebra lineage cashing out: the generator is the operator, Bochner is its curvature.

The obstruction (same shape). Bisimulation metrics are well-defined via a *discount* $c \in (0, 1)$ making the functional a contraction (Banach). But discounting shrinks distance geometrically with depth, pushing the geometry toward **ultrametric** (self-similar, totally disconnected) — and ultrametric spaces have *degenerate curvature*. The contraction that gives the metric fights the geodesic structure that gives curvature. **Resolution (the recurring move): discount the temporal direction ($::$), not the spatial one (near).** This is where irreversibility belongs, it leaves the K -metric undiscounted and able to be geodesic, and the discount factor becomes a **redshift / proper-time attenuation**. Whether the undiscounted spatial fixed point exists is the

technical crux.

Where the geometry lives. Bisimulation is sound only at the *real* layer (post-decoherence, branching-time). So the construction yields geometry on the real-weighted ciGSLT: **classical GR, by construction** — correct, since spacetime is a decohered, classical phenomenon. Geometry at the *complex* layer would need a metric on interference / linear-time, the object we established has no good branching pseudometric. **So “why is quantising gravity hard” gets a calculus- native answer: the metric apparatus is intrinsically a real-layer, branching-time construct, and the complex layer where it would have to live is the one without a behavioural metric.** The clean, reachable target is classical Einstein at the real layer via bisimulation- W_2 displacement convexity; the hard residue sits exactly where the whole arc says it must — at the boundary the quote crosses.

8. What to settle first

In rough dependency order:

1. **The phlo-potential’s shape** (§5). Convex $\Rightarrow$ explicit ϕ -dependence (a duller characterisation); non-convex/degenerate $\Rightarrow$ a genuine Higgs phase, the scale-invariant fixed point, the operator-weighted middle. The energetics framework is what can decide it. This gates the “phase” language everywhere.
 2. **The near^{real} domain = commutativity** (§6b, condition 2). Pinning this turns carried partiality into the quantum content and forces the dagger pair and the token-crossing.
 3. **The dual computes** $\text{compute}_R \dashv^\dagger \text{compute}_C$ (§6a). Without genuinely different (lossy vs reversible) computes the scalar colouring is cosmetic and the tower collapses to plain rho.
 4. **The quantum cost endofunctor** = $\text{decohere} \circ \text{CPM}$ (§3). Whether open-system evolution composes from two functors in the cost idiom, with the RG flow as iteration and the fixed point in the operator-weighted middle.
 5. **Wells over the continuum** (§4). A congruence on the $\mathbb{C}$ -state continuum is the prerequisite to *analysing* (not dissolving) the sharp Wigner’s friend — comparing two cuts rather than choosing between them.
 6. **The undiscounted spatial fixed point** (§7). Discounting time not space is the move; its existence is the crux for the whole GR programme.
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9. References introduced here (to verify)

Several are already in the companion note’s bibliography; the following are the new anchors of this line of thinking. Exact bibliographic details should be confirmed before use.

Diagrams, composition, and the dagger:

- J. Baez and M. Stay. *Physics, Topology, Logic and Computation: A Rosetta Stone*. In B. Coecke, ed., *New Structures for Physics*, Lecture Notes in Physics 813, pp. 95–172. Springer, 2011. (arXiv:0903.0340.) — the canonical “diagram = composite of cuts; evaluation = cut elimination.”
- P. Selinger. *Dagger compact closed categories and completely positive maps*. Electronic Notes in Theoretical Computer Science 170:139–163, 2007. — the CPM (doubling) construction;

the operator-weighted middle.

- B. Coecke and A. Kissinger. *Picturing Quantum Processes*. Cambridge University Press, 2017. — doubling-then-decoherence as a diagrammatic hierarchy.

Wigner’s friend / observer no-go:

- D. Frauchiger and R. Renner. *Quantum theory cannot consistently describe the use of itself*. Nature Communications 9:3711, 2018.
- K.-W. Bong et al. *A strong no-go theorem on the Wigner’s friend paradox*. Nature Physics 16:1199–1205, 2020. (Local Friendliness.)
- Č. Brukner. *A no-go theorem for observer-independent facts*. Entropy 20(5):350, 2018.
- B. Englert. *Fringe visibility and which-way information: an inequality*. Physical Review Letters 77:2154–2157, 1996. — the distinguishability/visibility bound.

Behavioural metrics and synthetic curvature:

- J. Desharnais, V. Gupta, R. Jagadeesan, P. Panangaden. *Metrics for labelled Markov processes*. Theoretical Computer Science 318:323–354, 2004.
- F. Otto. *The geometry of dissipative evolution equations: the porous medium equation*. Communications in PDE 26:101–174, 2001. — W_2 Riemannian structure.
- J. Lott and C. Villani. *Ricci curvature for metric-measure spaces via optimal transport*. Annals of Mathematics 169:903–991, 2009.
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- R. McCann. *Displacement convexity of Boltzmann’s entropy characterizes the strong energy condition from general relativity*. Cambridge Journal of Mathematics 8:609–681, 2020.
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- D. Bakry and M. Émery. *Diffusions hypercontractives*. Séminaire de Probabilités XIX, LNM 1123, pp. 177–206. Springer, 1985. — the Γ_2 criterion.
- M. Kunzinger and C. Sämann. *Lorentzian length spaces*. Annals of Global Analysis and Geometry 54:399–447, 2018. — causal/time-separation structure.
- R. Raussendorf and H. J. Briegel. *A one-way quantum computer*. Physical Review Letters 86:5188–5191, 2001. — adaptive measurement-based computation.

Already in the companion note (recalled here for the spectrum/congruence threads):

van Glabbeek (linear-time/branching-time spectrum); Larsen–Skou (probabilistic bisimulation); Ying (qCCS, bisimulation-as-congruence); Abramsky–Coecke (semiring of scalars, Born rule); Gell-Mann–Hartle / Griffiths / Omnès (consistent histories); Litvinov–Maslov (tropical dequantisation).